

Cycle-Compelling Colorings of Graphs

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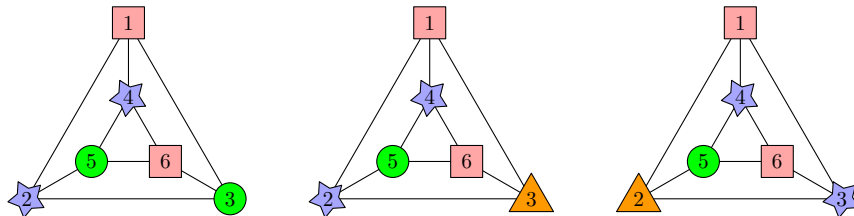
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Abstract. We define a cycle-compelling coloring of a graph as a proper coloring of the vertices such that every subgraph induced by one vertex of each color contains a cycle. The cycle-compelling number is defined to be the minimum k such that some k -coloring is cycle-compelling. We provide some general bounds and algorithmic results on this and related parameters. We also investigate the value in specific graph families including cubic graphs, disjoint union of cliques, and outerplanar graphs.

1 Introduction

Consider a graph G where the vertices are colored. A **rainbow committee** (RC) is a set consisting of one vertex of each color. We define a proper coloring of graph G as **cycle-compelling** if the subgraph induced by every RC of G contains a cycle. (Such a coloring exists if and only if G contains a cycle.) Then the **cycle-compelling number** of graph G , denoted $CCN(G)$, is the minimum number of colors needed for the existence of a proper coloring of G that is cycle-compelling. Further, the **guaranteed cycle-compelling number**, denoted $\mathcal{GCCN}(G)$, is the minimum number of colors such that every proper coloring with that many colors is cycle-compelling.

For example, let P be the prism of K_3 . Below are three colorings of P : the first two are not cycle-compelling and the rightmost one is cycle-compelling. The 3-coloring is unique up to isomorphism. And it is easy to see that every 5 vertices contain a cycle. Thus, $CCN(P) = 4$ and $\mathcal{GCCN}(P) = 5$.



The cycle-compelling number is a special case of the compelling number from [1]. Let $\mathcal{P}$ denote some property of subsets of vertices. Then a proper coloring of G **compels** $\mathcal{P}$ if every RC of G has property $\mathcal{P}$. Such a coloring might not exist, but if it does, then the **$\mathcal{P}$ -compelling chromatic number** of the graph G is the minimum number of colors in a proper coloring that compels property $\mathcal{P}$. For example we showed that the dominator chromatic number and total chromatic number correspond to $\mathcal{P}$ -compelling chromatic numbers for suitable choice of property $\mathcal{P}$. (For definitions of these parameters, see the recent survey [2].)

In Section 2 we provide some general bounds and facts, including characterizing the graphs where the cycle-compelling number is 3. Then in Section 3 we look at the parameters in graph families including joins, disjoint union of cliques, cubic graphs, and maximal outerplanar graphs.

2 Some General Bounds and Facts

It is trivial that CCN of the complete graph K_n is n , as it is for the cycle C_n . Likewise $\mathcal{GCCN}$ is n for these two graphs. Note too that if there exists a cycle-compelling coloring with k colors and k is less than the order, then there also exists one with $k + 1$ colors: simply choose any vertex that does not have a unique color and recolor it with a new hue. However, it is unclear whether all k -colorings being cycle-compelling implies all $(k + 1)$ -colorings are too.

As another example consider the complete bipartite graph $K_{r,s}$ with $r \geq s \geq 2$. The CCN is 4: For a suitable coloring, color one partite set with red and green, and color the other partite set with blue and orange. But $\mathcal{GCCN} = r + 2$. For a coloring with $r + 1$ colors that is not cycle-compelling, give the same color to all vertices in the smaller partite set and give a unique color to each vertex in the bigger partite set: every RC induces a star.

Recall that the **girth** $g(G)$ is the length of the shortest cycle of the graph G . The following result is a special case of Proposition 1 from [1].

Proposition 1 *Let G be a graph.*

- (a) $CCN(G) \geq \max\{g(G), \chi(G)\}$.
- (b) $CCN(G) \leq g(G) + \chi(G)$.

(The chromatic number is a lower bound since one needs a proper coloring; the girth is a lower bound since one needs the RC to contain a cycle. The chromatic number plus the girth is an upper bound since one can pick a shortest cycle and give each vertex therein a unique color and then color the remainder of the graph properly with new colors.)

These bounds can be achieved. For example, both the cycle and the complete graph achieve the lower bound; many disjoint union of cliques achieve the upper bound (see Theorem 11).

It follows that the smallest cycle-compelling number is 3. We determine next the graphs that achieve this:

Theorem 2 *For graph G , it holds that $CCN(G) = 3$ if and only if G is a complete tripartite graph.*

PROOF. Assume that $CCN(G) = 3$. Form an RC by picking one vertex of each color. The three vertices must form a cycle. It follows that vertices of different colors are always adjacent. That is, G is a complete tripartite graph. QED

On the other hand, there is no simple characterization of the case that $CCN(G) = 5$. Consider the join of K_2 with a graph H . In such a graph every proper coloring is cycle-compelling. So, since testing H for 3-colorability is

NP-hard, it follows that in general testing whether $CCN(G) = 5$ is NP-hard. It is unclear what happens for $CCN(G)$ equal to 4.

We now determine when the cycle-compelling number equals the girth:

Theorem 3 *If graph G has girth 4, then $CCN(G) = 4$ if and only if G is a complete bipartite graph.*

PROOF. We noted earlier that a cyclic complete bipartite graph has $CCN = 4$.

Assume $CCN(G) = 4$. Pick a vertex of each color. By the girth constraint, this RC forms a chordless 4-cycle, say $abcd$. Say a has color A , b has color B , c has color C and d has color D . Let a' be any other vertex; say it has color A . Then consider the RC $\{a', b, c, d\}$. This subgraph must contain a cycle, and by the girth condition it must be a 4-cycle. Vertex b is not adjacent to d , so b must be adjacent to a' . Similarly vertex d must be adjacent to a' . And by the girth, vertex a' is not adjacent to c .

Now consider any other vertex b' of color B . Start with the RC $\{a, b', c, d\}$. By the above reasoning, b' is adjacent to a and c but not d . So consider the RC $\{a', b', c, d\}$. By the same reasoning, a' is adjacent to b' and to d . That is, every vertex of color A is adjacent to every vertex of color B . Similarly this holds for every consecutive pair of colors on the original cycle. Further, a vertex of color B is not adjacent to any vertex of color D , while a vertex of color A is not adjacent to any vertex of color C . It follows that the graph is complete bipartite with bipartition $[A \cup C, B \cup D]$. QED

Theorem 4 *If graph G has girth at least 5, then $CCN(G) = g(G)$ if and only if G is a cycle.*

PROOF. We noted earlier that the cycle has CCN equal to its order.

Assume $CCN(G) = g(G)$. Pick a vertex of each color. This RC forms a g -cycle, say $abcd \dots za$, necessarily chordless. Suppose there exists another vertex; say a' has the same color as a . Then consider the g -tuple $\{a', b, c, d, \dots, z\}$. This subgraph must contain a cycle. By the girth condition, this must in fact form a chordless g -cycle. Vertex b is not adjacent to anything else; so it must be adjacent to a' . Similarly, vertex z must be adjacent to a' . Thus edges ab , az , $a'b$, and $a'z$ exist: but these edges form a 4-cycle, which is a contradiction. That is, vertex a' does not exist. QED

Turning to the guaranteed cycle-compelling number, let $LIF(G)$ denote the largest order of an acyclic subgraph, also known as the **largest induced forest**. (See for example [4].) If one uses more colors than this parameter, an RC has more vertices than this, and is therefore cyclic. Thus we obtain the following bound:

Proposition 5 *For all graphs G it holds that $\mathcal{GCCN}(G) \leq \max\{\chi(G), LIF(G) + 1\}$.*

A useful case of equality is the following result (using $\Delta(G)$ for maximum degree):

Observation 6 *If $\Delta(G) < \text{LIF}(G)$, then $\mathcal{GCCN}(G) = \text{LIF}(G) + 1$.*

PROOF. We know from Proposition 5 (and Brooks' Theorem) that $\mathcal{GCCN}(G) \leq \text{LIF}(G) + 1$. Let J be a largest induced forest. Give each vertex of J a different color. Then extend to a coloring of G using these colors (which can be done greedily because every vertex has fewer neighbors than available colors). The resultant coloring uses $\text{LIF}(G)$ colors and the set J is an RC that is acyclic. QED

In contrast, an example of a graph where the bound of Proposition 5 is not attained is a complete tripartite graph.

One can also consider graphs with large values of the chromatic number. For example:

Theorem 7 *If graph G of order n has chromatic number at least $(n+3)/2$, then $\text{CCN}(G) = \mathcal{GCCN}(G) = \chi(G)$.*

PROOF. Consider any proper coloring using the minimum number of colors. By the pigeonhole principle there must be at least three colors that are used only once. Since each of these colors is necessary, these vertices must be mutually adjacent. That is, every RC contains a triangle. QED

3 Graph Families and Operations

3.1 Join

We next consider the join of two graphs. As in [1], we define a proper coloring of a graph as **edge-compelling** if every RC contains an edge, and the **edge-compelling number** as the minimum number of colors in an edge-compelling coloring. We start with the join of a graph with a single vertex:

Theorem 8 *For any graph H , it holds that $\text{CCN}(H \vee K_1)$ is 1 more than the edge-compelling number of H .*

PROOF. Every proper coloring of $H \vee K_1$ gives the universal vertex v a unique color and H a proper coloring. It follows that every RC J of $H \vee K_1$ contains v , and so J induces a cyclic subgraph if and only if $J - \{v\}$ is not independent. QED

A similar result holds for the guaranteed cycle-compelling number. On the other hand, consider the case that graphs G and H both contain edges. Then every proper coloring of the join is cycle-compelling, since every RC contains (at least) two vertices from G and (at least) two vertices from H and these four vertices contain a 4-cycle. Thus:

Observation 9 *For any graphs G and H that contain at least one edge, it holds that $\text{CCN}(G \vee H) = \mathcal{GCCN}(G \vee H) = \chi(G) + \chi(H)$.*

3.2 Disjoint union of cliques

Turning to disjoint union, one can write bounds for CCN of the union in terms of the parameters of the components, but these are mostly no better than the bounds given in Proposition 1. One improvement that does arise is that $CCN(G \cup H) \leq CCN(G) + CCN(H) - 1$, since an RC is still compelling if two components share one color.

For specific graphs we consider the case where the components are complete. Assume graph G is the disjoint union of cliques with maximum clique order m . By Proposition 1, it holds that $m \leq CCN(G) \leq m + 3$. Here are two instances where the lower bound is attained:

Observation 10 *Let G be the disjoint union of s cliques with the largest clique of order m and the second-largest clique of order ℓ .*

- (a) *If $m \geq \ell + 3$, then $CCN(G) = m$.*
- (b) *If $m > 2s$, then $CCN(G) = \mathcal{G}CCN(G) = m$.*

PROOF. (a) Consider a proper $(\ell + 3)$ -coloring such that all but the largest clique are colored using the same palette of ℓ colors. Then there are (at least) three colors that appear only on the biggest clique, and so the corresponding vertices form a triangle in every RC.

(b) Consider any proper coloring. By the pigeonhole principle, every RC must contain at least 3 vertices from some clique. QED

First we resolve the case of two cliques:

Theorem 11 *Consider the disjoint union $K_\ell \cup K_m$ with $3 \leq \ell \leq m$. Then $CCN = \mathcal{G}CCN = \max\{m, 5\}$.*

PROOF. If $m \geq 5$, then the result follows from part b of Observation 10. So assume $m \leq 4$. Every 5-coloring works by the pigeonhole principle.

Consider a coloring of the graph with four colors. Then we claim there exists an RC J consisting of two vertices from each clique and therefore acyclic. Begin constructing J by taking up to two vertices with unique colors from each clique; then try to complete to the claimed RC. Suppose one gets stuck. Since each clique has at least three colors, we must have two vertices from one clique and one vertex from the other, and the fourth color exists only in the first clique. But then by the way the RC was started, the first clique contains three vertices of unique colors, which does not leave enough colors for the second clique to be colored, a contradiction. QED

Second we resolve the case where the cliques have the same order: To show that a coloring is not compelling, we use the well-known weighted version of Hall's theorem (see for example [3]).

Theorem 12 *Consider a collection $\mathcal{X} = (X_1, \dots, X_s)$ of sets where each X_i has a value b_i . There exist disjoint sets $B_i \subseteq X_i$ with $|B_i| = b_i$ if and only if for all subsets D of $\{1, \dots, s\}$ it holds that $|\bigcup_{i \in D} X_i| \geq \sum_{i \in D} b_i$.*

In our application, we write the values b_i as a vector $\mathbf{b}$.

Theorem 13 *Consider the disjoint union sK_m for $m \geq 3$. The cycle compelling number is:*

$$\begin{aligned} m & \quad \text{if } s < \lceil m/2 \rceil, \\ 2s + 1 & \text{ if } s = \lceil m/2 \rceil, \text{ and} \\ m + 3 & \text{ if } s > \lceil m/2 \rceil. \end{aligned}$$

PROOF. For the upper bound: $m + 3$ is always an upper bound, as was observed above. Similarly, any coloring using at least $2s + 1$ colors is automatically compelling by the pigeonhole principle.

For the lower bound: m is always a lower bound on the chromatic number. Assume $s > \lceil m/2 \rceil$ and consider a coloring with $m + 2$ colors. We show that there exists an acyclic RC. Create family $\mathcal{X}$ of size $\lceil m/2 \rceil + 1$ as follows: use as sets (the colors in) the first $\lceil m/2 \rceil$ cliques and then combine the colors in the remaining cliques together as one set. Assume first that m is even, and let $\mathbf{b} = (2, 2, \dots, 2)$. Consider a sub-collection $\mathcal{T}$ of $\mathcal{X}$ containing t sets. Then $\mathcal{T}$ has at least m colors, which is at least $2t$ except if $t > m/2$. That is, we may assume that $t = m/2 + 1$; but that means $\mathcal{T} = \mathcal{X}$ and so all $m + 2$ colors are present. Hence by Theorem 12 there is an RC with two vertices from each clique.

So assume that m is odd, and let $\mathbf{b} = (2, 2, \dots, 2, 1)$. Again the condition on $\mathcal{T}$ is immediate except for $t > m/2$. If $t = (m + 3)/2$, then $\mathcal{T}$ is all of $\mathcal{X}$ and has all $m + 2 = 2t - 1$ colors. So assume $t = (m + 1)/2$. Then the condition on collection $\mathcal{T}$ is satisfied unless $\mathcal{T}$ is the first $(m + 1)/2$ cliques and all of these cliques have the same $m = 2t - 1$ colors. But since more than m colors are used in total, we can go back and index $\mathcal{X}$ such that the first and second clique do not have exactly the same colors. Hence by Theorem 12 there is an acyclic RC.

Assume now that $s = \lceil m/2 \rceil$. Then consider a coloring with $2s$ colors. One can in a very similar way apply the above theorem with $(2, 2, \dots, 2)$. We omit the details. QED

By the above result and Observation 6 one can deduce the following:

Theorem 14 *For $m \geq 3$ it holds that $\mathcal{GCCN}(sK_m) = m$ if $s < \lceil m/2 \rceil$ and $\mathcal{GCCN}(sK_m) = 2s + 1$ otherwise.*

3.3 Cubic Graphs

By Theorem 2, an r -regular graph G with $CCN(G) = 3$ is a complete tripartite graph with equal number of vertices in each partite set. So it follows that r is even. That is, there is no cubic graph G with $CCN(G) = 3$.

There are a few cubic graphs with CCN equal to 4. Namely, K_4 , $K_{3,3}$ and P , the prism-of- K_3 , discussed earlier. These are all the examples, as we now show.

Theorem 15 *There is no cubic graph G of order 8 or more with $CCN(G) = 4$.*

PROOF. By Theorem 3, the only cubic example with girth 4 is $K_{3,3}$. So we may assume G has girth 3 and order at least 8.

Consider a triangle T of G with vertices a, b, c . Let d be any vertex of the fourth color. Suppose vertex d is not adjacent to T . Consider any neighbor e of d . One can build an RC with d and e and two vertices of T . The only way this RC can contain a cycle is that e is adjacent to the two vertices taken from T . But this has to be true for each choice of e , and thus there are six edges going into T , but only three coming out, a contradiction. Thus vertex d is adjacent to T .

Consider some vertex f that is not adjacent to T . (Such a vertex exists since G has order at least 8.) We may assume that f has the same color as a . We can build an RC with vertices d, f, b , and c . The only way this RC can contain a cycle is that d is adjacent to b and c . There can only be one vertex with two neighbors in T ; thus d is the unique vertex of the fourth color.

By the same argument applied to the triangle $\{b, c, d\}$, it follows that a is the unique vertex of its color. This contradicts the fact that a has the same color as f . That is, there is no such G . QED

In contrast, there are infinitely many cubic graphs G with $CCN(G) = 5$. For example, take a bipartite cubic graph and perform a Δ - Y operation at one vertex v : that is, replace vertex v with a triangle $v_1v_2v_3$ with the same neighbors. Then a cycle-compelling coloring is obtained by using three colors on the triangle and two colors on the remainder (which is bipartite).

It is easy to see that every cubic graph G of order at least 6 has an induced forest of order at least 4. Thus by Observation 6 for these it holds that $GCCN(G) = LIF(G) + 1$.

3.4 Maximal Outerplanar Graphs

Recall that a **maximal outerplanar graph** is obtained from a cycle drawn in the plane by adding chords inside to make a triangulation. For a maximal outerplanar graph G , it follows from Proposition 1 that $CCN(G) \leq 6$, since the girth and chromatic number are both 3. According to computer, the snake P_9^2 has $CCN = 6$. From this it can be shown that all longer snakes have $CCN = 6$ also. On the other hand, by Theorem 8 and the value of the edge-compelling number of the path given in [1], the fan $F_n = K_1 \vee P_{n-1}$ has $CCN = 5$ for $n \geq 8$.

Finally we consider the guaranteed cycle-compelling number. There are many cases where the parameter equals the upper bound from Proposition 5. If the degree is small, such as in the snake P_n^2 , this follows from Observation 6. But even if the degree is large, such as the fan F_n , the result can be shown. We omit the details.

4 Future Thoughts

Open questions already mentioned include resolving the value of CCN for all disjoint union of cliques. One intriguing question is whether one can duplicate a

vertex and have the cycle-compelling number go down. Other questions include considering the parameters in graph families such as products. In another direction, one could also consider colorings where there exists an RC that contains a cycle.

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References

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